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Evolution of the Queen Charlotte Basin; further paleomagnetic evidence of Tertiary extension and tilting

E. Irving [a,*], J. Baker [a], P.J. Wynne [a], T.S. Hamilton [b], M.T.D. Wingate [c]

[a] *Pacific Geoscience Centre, P.O. Box 6000, Geological Survey of Canada, Sidney B.C. V8L 4B2, Canada*
[b] *Hamilton Geoscience Services, 2180B Calvin Ave, Sidney B.C. V8L 4C3, Canada*
[c] *Tectonics Special Research Centre, Department of Geology and Geophysics, The University of Western Australia, Nedlands W.A. 6907, Australia*

Abstract

The Queen Charlotte Islands form the northwestern margin of the Queen Charlotte Basin. The basin, which was formed by extension, contains numerous mid-Tertiary lavas, plutons and dykes, referred to as the Masset Igneous Complex, and Eocene to Present marine and non-marine sedimentary rocks. Throughout its history, the basin has been bounded on the west by a north-northwest oriented transform plate boundary, the Queen Charlotte Fault. Initially, motion along this fault was transtensional (phase 1), then more purely transform (phase 2), and finally transpressive (phase 3). Most of the Masset Igneous Complex appears to have been formed under transtension (phase 1, Middle Eocene through Early Miocene). Previous paleomagnetic studies of the complex (129 sites mainly from the southern Queen Charlotte Islands) and new results herein (102 sites mainly from the northern islands) illustrate magnetizations with inclinations generally steeper than those expected from coeval rocks of continental North America, indicating north–south extension, segmentation and tilting of fault-bound slivers along the western margin of the basin. We argue that this north–south extension occurred in phase 2, during the Middle and/or Late Miocene, long after the main east–west extension of phase 1. Results from the northwest corner of the Queen Charlotte Islands, although commonly anomalous, are not oversteepened, and may delineate a structural domain separate from the rest of the archipelago.

As a by-product, this study provides two reference paleopoles for North America – one for the Late Eocene–Early Oligocene from Ramsay Island lavas (78.1°N, 119.9°E, $A^{\circ}_{95}=7.0^{\circ}$), and one of lesser reliability for the Early Miocene from the lavas of Dinan Bay (83.3°N,137.8°E, $A^{\circ}_{95}=9.8^{\circ}$). A second by-product is that the magnetic polarities and radiometric ages are consistent with the Cande and Kent global geomagnetic polarity time scale. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: extension; paleomagnetism; Queen Charlotte Basin; tilting

1. Preamble

First of all we wish to express our appreciation of Myrl Beck's imaginative and lively approach to science, and to say how pleased we are to have our paper included in his commemorative volume. Myrl and his colleagues (Diehl et al., 1983) were responsible for many of the results from which the reference fields, so basic to this paper, were calculated.

Many studies of paleomagnetism have yielded

* Corresponding author.

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anomalous results that were inconsistent with contemporary ideas about structure and tectonic history, and it has long been recognized that previously undetected tilts sometimes may be responsible (e.g. Hood, 1961). In the Cordillera of North America, it was Beck and Noson (1972) along with Lecouter and Ager (1972), who first recognized that tilting was a possible explanation for anomalous results that otherwise required displacements of large parts of the Cordillera relative to cratonic North America (Beck et al., 1981; Beck, 1991). Most students of the tectonics of the Cordillera, at some stage in their investigations, have found themselves confronted by Myrl's question, 'tilt or translation?', arising either from their own paleomagnetic work or from that of others.

During the past few years, some detailed answers to Myrl's question have been given. Some paleomagnetic anomalies are best explained by tilting (e.g. Wingate and Irving, 1994). Some seem to require large displacements (e.g. Wynne et al., 1995; Ague and Brandon, 1996). Others may require both (Irving et al., 1994). In this paper we are concerned with anomalous directions of magnetization from mid-Tertiary rocks of the Queen Charlotte Islands, which are an example of the former process, and which provide evidence for regionally consistent tilting on a scale not previously suspected.

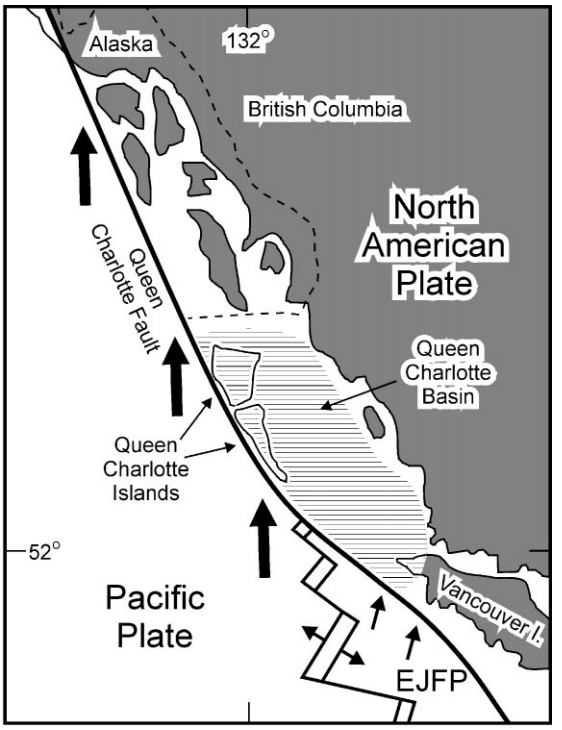


Fig. 1. Present tectonic setting of the Queen Charlotte Islands. The Pacific Plate is converging highly obliquely (bold arrows) with the North American Plate, as it has done since the Late Miocene. EJFP Explorer – Juan de Fuca Plate.

2. Introduction

The Queen Charlotte Basin (Fig. 1) was defined by Shouldice (1971). It originated in the Eocene. Summaries of its tectonic history have been given by Yorath and Chase (1981), Yorath and Hyndman (1983) and Hyndman and Hamilton (1993), and of its sedimentation history by Higgs (1991). The basin occupies the Queen Charlotte Islands, Dixon Entrance, Hecate Strait and Queen Charlotte Sound (Higgs, 1988, 1991; Thompson, et al., 1991). It is bounded on the west by the Queen Charlotte Fault, the transform boundary between the Pacific and North American plates. Its other boundaries are less distinct, and can be taken to follow roughly the coasts of Alaska and British Columbia, except in the south, where the basin occupies northern Vancouver Island as far south as the Brooks Peninsula (Lewis et al., 1997). Important structural features are shown in Fig. 2. The basin is underlain by Mesozoic rocks, which extend beyond its present limits, and which accumulated under a different plate tectonic regime.

According to Hyndman and Hamilton (1993), the basin was formed as a result of the establishment of the Queen Charlotte Fault in the Eocene, and its subsequent evolution was intimately related to the evolution of this plate boundary. From various lines of evidence, they argued that the basin could now be about one and a half times wider than it was originally (150% extension).

According to Higgs (1991) the basin-fill comprises two main sequences reflecting a two stage subsidence history of the type described by McKenzie (1978). The oldest is Middle Eocene through Lower Miocene (Unit 1), which is a diachronous rift succession of igneous and sedi-

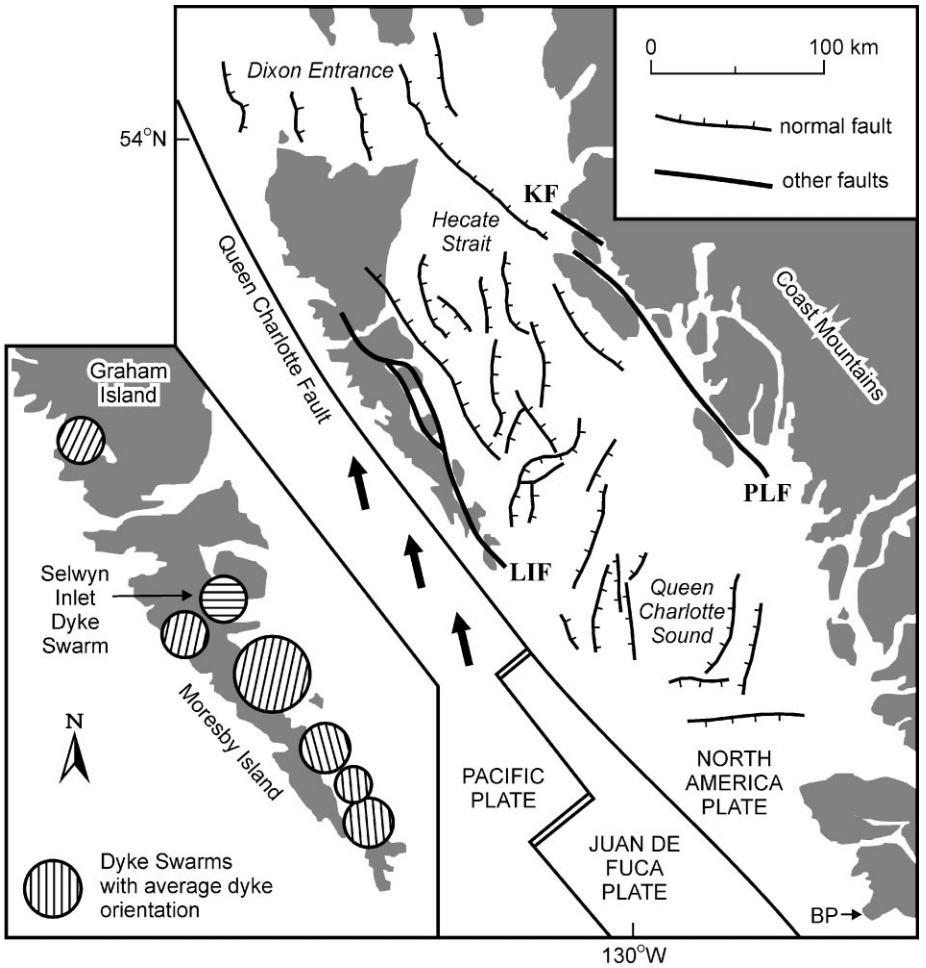


Fig. 2. Some structural features of the Queen Charlotte Basin. BP, Brooks Peninsula; KF, Kitkatla Fault; LIF, Louscoone Inlet fault system; PLF, Principe–Laredo Fault. Lower left, the predominant directions of dykes in the large swarms (Souther and Jessop, 1991) are shown; all have been surveyed paleomagnetically. Compiled from Sutherland Brown (1968), Lewis and Ross (1991), Woodsworth (1991), Rohr and Dietrich (1992), Rohr and Currie (1997).

mentary rocks, deposited in half-grabens. Offshore, Rohr and Dietrich (1992) have mapped these half-grabens (Fig. 2). Rohr and Dietrich (1992) and Rohr and Currie (1997) have suggested ways in which rifting might have occurred. Igneous rocks associated with this initial rift phase, include volcanics, small plutons and dykes, and are the focus of this study (Fig. 3). A well documented feature of the dykes is that, except for one younger swarm, they are oriented on average north–south, indicating that they were intruded under a regime of generally east–west extension (Souther and Jessop, 1991).

Following this rift succession, the basin was blanketed by Miocene to Recent sedimentary strata (Unit 2 of Higgs, 1991) deposited during regional subsidence, which occurred presumably as a result of cooling. Post-rift Unit 2 is absent from most of the western side of the Queen Charlotte Islands and from Dixon Entrance. The seismic work of Rohr and Dietrich (1992) has confirmed this bipartite structure of the basin.

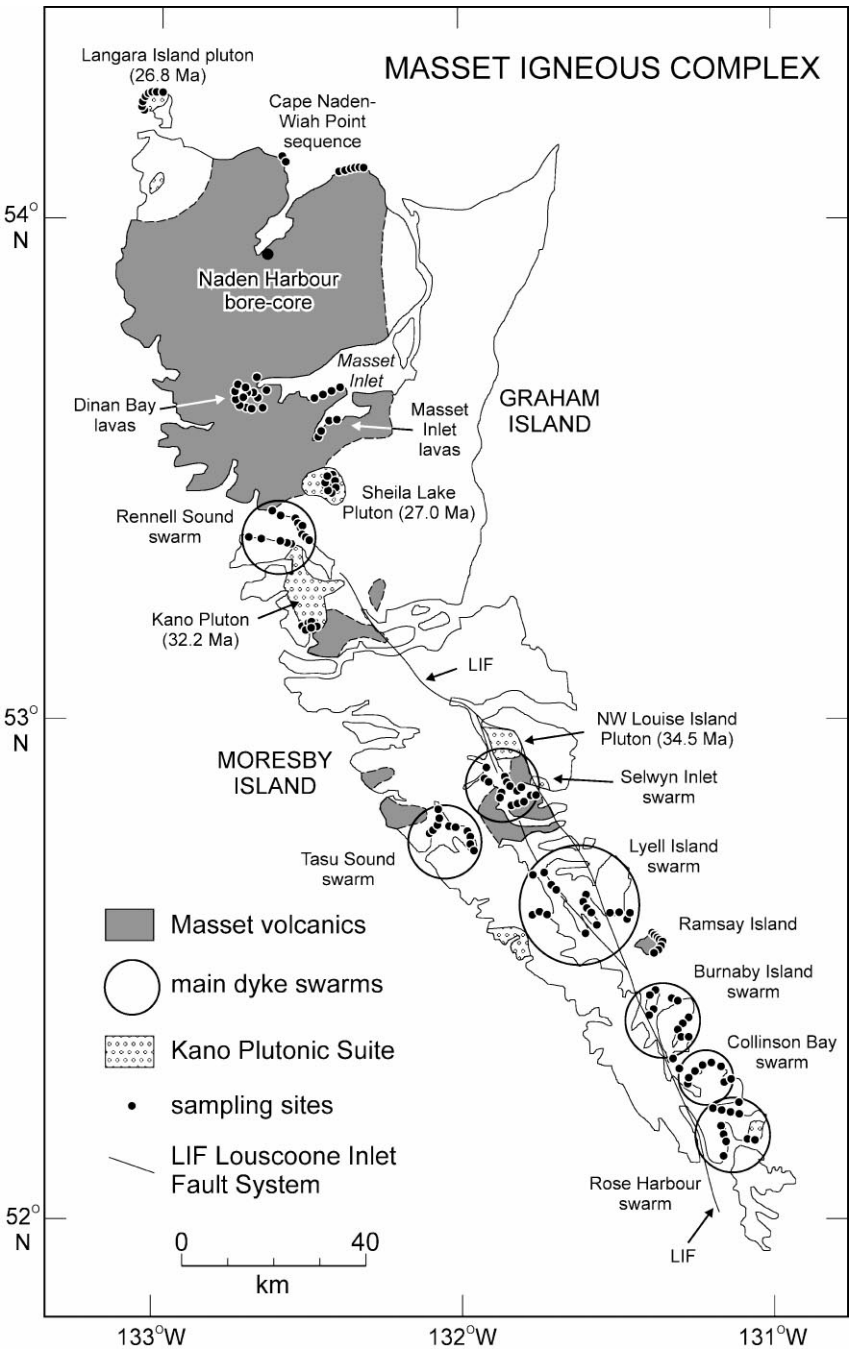


Fig. 3. The Masset Igneous Complex, rock units studied, and sampling sites. LIF, Looscoone Inlet fault system. Compiled from Sutherland Brown (1968), Anderson and Reichenbach (1991), Hickson (1991), Souther and Jessop (1991), Irving et al. (1992), Hamilton and Dostal (1993) and present study.

The mid-Tertiary igneous rocks, which are a characteristic feature of the Queen Charlotte Basin, are referred to collectively as the Masset Igneous Complex. Previous paleomagnetic studies of the complex, mainly in the southern half of the archipelago, found that the inclinations of remanent magnetizations are about 10° steeper than those expected from coeval rocks from continental North America (Irving et al., 1992). As an explanation, these authors proposed that the archipelago has undergone north–south extension and was fragmented into blocks which were then tilted 9° to16° down to the north. The proposed north–south extension is consistent with the structural analysis of Lewis (1991) and Lewis and Ross (1991), who recognized that a limited area of central Moresby Island and adjacent islands, situated east of the Louscoone Inlet fault system (Sutherland Brown, 1968, Fig. 2) had undergone fragmentation into northerly-tilted fault-blocks.

Evidently, there was in the Queen Charlotte Islands an initial phase of east–west extension followed by a phase of north–south extension. Irving et al. (1992) proposed two sequential phases of north–south extension, but new evidence, described below, requires only one.

In this paper, we present (1) new paleomagnetic data from mid-Tertiary igneous rocks mainly from northern Queen Charlotte Islands, which provide further examples of oversteepened magnetizations, (2) a test that verifies the paleomagnetic method of estimating tilt, (3) evidence that extension and tilting probably occurred in the Middle through Late Miocene, and (4) we integrate our results into a tectonic model of basin evolution, based mainly on the work of Rohr and Dietrich (1992) and Hyndman and Hamilton (1993). This study is a geographical extension of the work of Irving et al. (1992).

3. Masset Igneous Complex (MIC) and tectonic setting

The Masset Igneous Complex (MIC, Fig. 3) consists of basaltic and andesitic lavas, dykes, and small, monzonite or diorite plutons (Sutherland Brown, 1968; Yorath and Chase, 1981; Souther and Bakker, 1988; Anderson and Riechenbach, 1991; Hickson, 1991; Souther and Jessop, 1991; Hamilton and Dostal, 1993). Igneous activity began in the Middle Eocene, continuing for about 25 Ma through Early Miocene, and sporadically into the Middle Miocene. By Late Miocene time (ca 10 Ma), it had ceased. Except for the Middle Miocene Selwyn Inlet dyke swarm (hornblende K–Ar age 13 ± 6 Ma), the rocks that have been studied paleomagnetically range in age from Middle Eocene through Early Miocene

In the northern Queen Charlotte Islands, lavas are widespread (Fig. 3), and are typically Upper Oligocene through Lower Miocene (Hickson, 1991; Hamilton and Dostal, 1993). In the south, there are isolated patches of Upper Eocene to Lower Oligocene lavas. Also in the south, Upper Eocene and Lower Oligocene plutons and dykes are common, and presumably fed formerly widespread lava fields, of which these isolated lava occurrences are remnants. Hyndman and Hamilton (1993), therefore, argued that volcanism was semi-continuous throughout the interval Middle Eocene to Early Miocene, and that the present irregular distribution of volcanics reflects, mainly, the greater erosion suffered by older volcanic strata.

Evidence from bore-cores (Hamilton and Dostal, 1993) and seismic surveys (Rohr and Dietrich, 1992) indicate that, beneath Queen Charlotte Sound and Hecate Strait, volcanics occur most commonly in half-grabens, and are probable equivalents of the MIC. Other probable equivalents also occur on the mainland east of Hecate Strait (Woodsworth, 1991).

While there is debate concerning the time scale of Tertiary magnetic anomalies and the interpretation of plate circuit calculations (Norton, 1995; Atwater and Stock, 1998), their is consensus on two questions – that prior to the later Middle Eocene (ca 45–42 Ma), the oceanic plate to the west was strongly convergent with North America, and that later, at ca 5 Ma (Late Miocene or Early Pliocene), plate motions became obliquely convergent, initiating a transpressive regime that continues to the present. There are uncertainties in the precise time at which these changes occurred, but these uncertainties are not critical for our

work. What is important is that the changes themselves, which affected the entire Pacific basin, are well resolved (Engebretson et al., 1985 Stock and Molnar, 1988; Hyndman and Hamilton, 1993; Atwater and Stock, 1998). Between ca 45 and ca 5 Ma (interval B of the synoptic diagram in Fig. 4), various plate reconstructions of motions along the Queen Charlotte Fault agree only to the extent that motions were dominantly transform; varying degrees of convergence and divergence are permitted, but they never significantly exceeded 15°, and were generally $<10°$. Clearly, however, the details of stress history during interval B have to be determined by means other than plate tectonics.

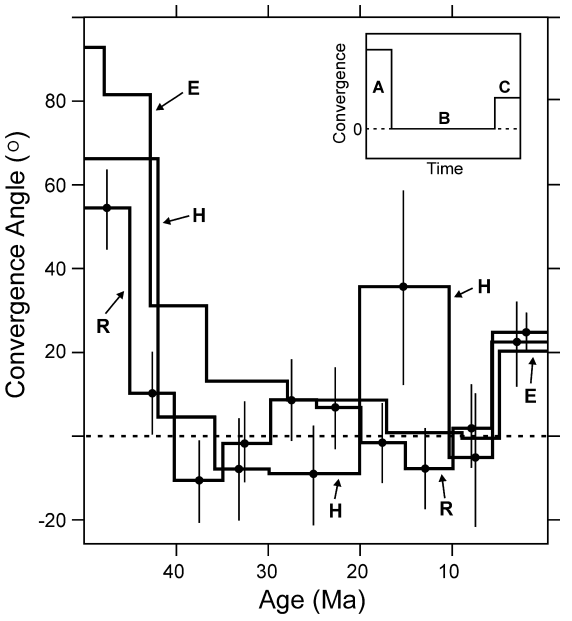


Fig. 4. Changes in direction of relative motion of Pacific and North American plates. The azimuth of their common margin is assumed to be aligned 40° west of north. E from Engebretson et al. (1985) has no error assignments. H calculations of Hyndman and Hamilton (1993) using rotations and errors given by Stock and Molnar (1988). R calculations of Rohr and Currie (1997) using rotations of Norton (1995) and an average error of 10°; note that the component of convergence between 20 and 10 Ma in H has a large error. The change at ca 45 Ma was accompanied by a large decrease in relative velocities. In the synoptic diagram, A and C are the well established intervals of strong convergence, and highly oblique convergence respectively. B is the interval of *predominantly* transform motion; initially transtensive, more nearly transform later (phases 1 and 2 respectively of Fig. 12).

The occurrence of abundant dykes (Souther and Jessop, 1991), of seismically identified half-grabens (Rohr and Dietrich, 1992), and the petrology and geochemistry of the lavas (Hamilton and Dostal, 1993), require an extensional environment and the generally north–south alignment of dykes and normal faults shows that extension was predominantly east–west (Fig. 2); the geological evidence for extension (Fig. 2), is compelling. Hyndman and Hamilton (1993), therefore, proposed that rifting of Queen Charlotte Basin and the formation of the MIC occurred under a transtensional regime.

4. Paleomagnetic estimation of tilt

The direction and amount of tilting at a test locality can be estimated by comparing its mean direction of magnetization with directions expected (reference field directions) from measurements made from coeval rocks in tectonically stable continental North America. Sampling at test localities must be sufficient to average out secular variation. The basic assumption is that the time averaged geomagnetic field is that of a geocentric dipole, a question we return to later.

If a test locality direction conforms to that expected, then it is very unlikely that the locality has been tilted. If it differs significantly, then the locality could have been tilted, but there may be other explanations, as described below.

Tilts are calculated by rotating the measured direction about a horizontal axis into that expected. Tilting solutions are not unique; any observed direction can be made to conform to any expected direction, either by a single rotation reflecting a single tilting event, or by a sequence of rotations reflecting several such events. We adopt the simplest solution, that is, a single rotation about a horizontal axis – the rotation that causes the directions to migrate along a small circle from the observed to the expected direction.

5. Paleomagnetic methods

Previous work showed good internal consistency within rock units (Irving et al., 1992,

Table 4), and that a good estimate of their mean directions could be obtained from 10 or so sampling sites. Accordingly, we aimed to obtain 10 or more sites from each pluton, lava sequence or dyke swarm, which are referred to as 'rock units'. For logistical reasons, we were not always able to do so. The reliability test, described below, of paleomagnetically determined tilts, was based on more detailed sampling.

Five to seven oriented cores were drilled at each site and oriented by sun and magnetic compass; usually two specimens were cut from each core. Measurements were made using automated Schonstedt spinner magnetometers (Irving et al., 1992). For analysis we used the computer programmes of Enkin (1991).

There are three types of behaviour. Ninety six sites out of a total of 102 have single stable magnetizations and Brunhes-age overprints of variable magnitude. The latter are directed along the present geomagnetic field, and are probably recently acquired viscous remanent magnetizations (VRM). They can readily be removed by heating to low temperatures or by treatment in low alternating fields (AF) in the absence of an ambient field. After removal of Brunhes overprints, orthogonal plots of AF and thermal demagnetization of pilot specimens show excellent linearity to the origin (Fig. 5) with the exception of some specimens from the Rennell Sound dyke swarm. Six sites have Brunhes overprints only, with no detectable underlying stable magnetizations. Finally, there is one site whose magnetization is unusually strong, and has properties comparable with isothermal remanent magnetization (IRM) as if struck by lightning. Brunhes overprints and lightning magnetizations (7% of collections) provide no information about the paleofield and are not reported.

In our earlier work, we showed that complex magnetizations often of opposed polarity, and carried by magnetite, pyrrhotite and hematite, are common in dykes, but absent in lavas or plutons (Irving et al., 1992). The magnetizations in this study are carried by magnetite with occasional minor contributions from hematite with the same direction. No evidence of pyrrhotite, or of complex

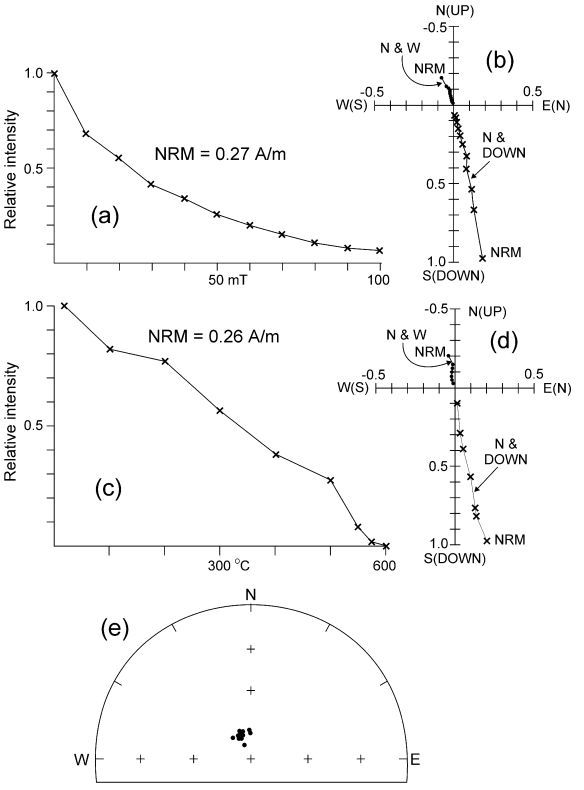


Fig. 5. Demagnetization of basalt from site 66, Ramsay Island. (a)–(d) Magnetization as a function of AF and thermal demagnetization of companion specimens from the same core; graphs of intensity, and orthogonal plots on the horizontal (dots) and vertical (crosses) north–south planes. (e) Directions of principal magnetization in 11 specimens relative to present horizontal.

magnetizations, was found, presumably because most samples were from lavas and plutons.

Demagnetization behaviour has been extensively documented (Irving et al., 1992), and some general remarks will suffice. In plutons, most pilot specimens yielded excellent square-shouldered thermal demagnetization curves with unblocking temperatures confined between 500 and 600°C, indicating that single domain magnetite is the carrier of the remanence, and Brunhes overprints are absent. About a third of the specimens from plutons show a decrease beginning at <300°C, as Brunhes overprints of variable magnitude are removed. In lavas, square shouldered 'magnetite' curves also occur, but more commonly the unblocking temperatures range from room temper-

ature more or less continuously up to 600°C, with small shoulders between 400 and 550°C. An example is shown in Fig. 5; below 300°C a Brunhes overprint is being demagnetized, and from 300 to 600°C a unidirectional magnetization is progressively removed. We interpret this to reflect the occurrence of magnetite with a wide range of grain sizes, the smallest carrying the highest temperature remanence. In both lavas and plutons, unblocking temperatures do not exceed 600°C, indicating that haematite is absent. In the Rennel Sound dyke swarm unblocking temperatures range from 400 to 600°C indicating that magnetite is the chief carrier, but in some specimens there are small magnetizations having unblocking temperatures of up to 650°C, presumably due to hematite.

Over 90% of specimens have magnetizations consisting of a single stable magnetization and a Brunhes overprint that can be readily removed by demagnetization. Within each lava sequence, pluton or dyke swarm the directions of natural remanent magnetization (NRM) were generally well grouped, and the directions in pilot specimens did not change significantly after removal of small overprints. For such simple magnetization, the question arises whether multiple demagnetization steps followed by principle component analysis should be performed for each specimen, or whether it is sufficient to carry out single-step demagnetization, that is, a single demagnetization at a level somewhat above that required to remove the Brunhes overprint, the level being determined from detailed demagnetization of pilot specimens from each site. The following tests show that, for most rocks of the present collection, the latter is justified, because there is little variation in demagnetization behaviour between specimens from the same site.

As a general check, we compiled directions from line-fitting of 58 pilot specimens, calculated means for each rock unit, and then compared them with means obtained by single-step demagnetization of all specimens. The average difference between means is 1.8°, and differences are always less than the combined standard errors ($P=0.67$). The two procedures, therefore, give statistically identical results.

We made another test, using data of Fig. 5. Least squares fit to decay lines gives the direction $D,I=341°,78°$ ($\alpha_{95}=3°$) for AF demagnetization from 10 to 100 mT, and 359°,80° ($\alpha_{95}=2°$) for thermal demagnetization between 100 and 580°C of the companion specimen from the same core. These directions are indistinguishable from one another. Also indistinguishable are measurements at any step within these demagnetization ranges, for example, the direction after AF demagnetization at 30 mT is 343°,79°, and the direction in the companion specimen after heating to 500°C is 339°,81°. There is no significant difference among results obtained by the two methods.

We made a third test, using data from 12 lava flows from Ramsay Island. Line-fitting in the range 10–40 mT and 100–550°C yielded a mean $D,I=$ 339.2°,80.7°, $N=12$, $k=43$, $\alpha_{95}=5.7°$. Single-step cleaning at 20 mT or 500°C yielded $D,I=$ 343.7°,80.9°, $k=37$, $\alpha_{95}=6.1°$. These are not significantly different; single-step demagnetization suffices.

Rennell Sound dykes are different in one minor respect. Demagnetization of most specimens shows linear decay to the origin, but specimens from a few sites have small intercepts on the axes of orthogonal plots. This was true for both AF and thermal demagnetization. As a safeguard, therefore, we demagnetized all specimens from the swarm at several steps and carried out principal component analysis. The mean from single-step demagnetization (11 sites) is $D,I=3.3°,83.1°$, $k=$ 102, $\alpha_{95}=4.5°$, and from line-fitting $D,I=$ 10.4°,81.9°, $k=103$, $\alpha_{95}=4.5°$. These are statistically identical, so the effect of the intercepts is small and random, hence single-step demagnetizations suffice. The effect is an artifact of laboratory processing. Detailed tests showed that specimens with intercepts had acquired, in the laboratory, small temporary VRMs: they were acquired at the same stage during processing and always in the same direction relative to the specimen. Specimen orientations are essentially random and so also is the effect of the intercepts, as the above test shows.

6. Test of paleomagnetic method of tilt estimation

To make this test, we studied an inclined lava sequence from Ramsay Island which is geographi-

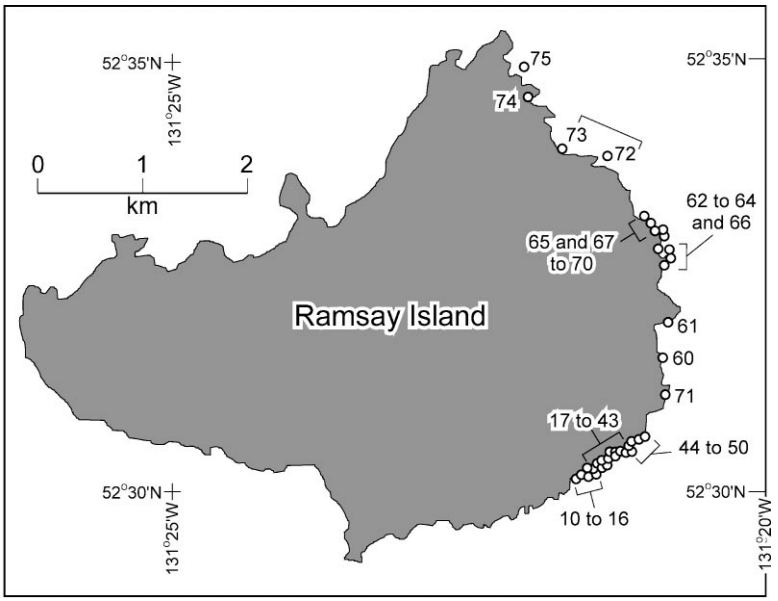


Fig. 6. Ramsay Island and sampling sites. Lavas have uniform northerly dips, younging to the north. Brackets show the arrangement of sites into 11 groups (some comprise a single sampling site) to provide roughly uniform coverage through the section, as explained in the text. General location in Fig. 3.

cally very close to previously studied coeval dykes (Figs. 3 and 6). This is one of the few localities in the Queen Charlotte Islands where there is a thick sequence of lavas with interleaved sediments from which bedding attitudes can be determined. The test is that if, relative to present horizontal, the directions of magnetization are the same, irrespective of sign, as those in adjacent dykes, and if, after correction for bedding, the directions in lavas agreed with continental reference directions, then it is very likely that both lavas and dykes have been tilted.

The rocks are andesites and basalts which have yielded K–Ar ages of 41–36 Ma, Late Eocene to Early Oligocene (Hamilton and Dostal, 1993). We sampled 34 lava flows and three dykes over a distance of 4 km. Attitude of bedding was measured at 29 places close to sampling sites, using interfaces between flows, interfaces between flows and layering in underlying beds of agglomerate or conglomerate, bedding in fine-grained sediments within conglomerate layers, and alignment of vesicles (footnote Table 1). Poles to bedding are well grouped, with the mean 172.9°,72.2°, $N=29$, $k=46$, $\alpha_{95}=4.0°$. Places at which measurements were made are not uniformly spaced stratigraphically, and this procedure gives too much weight to sections with limited stratigraphic coverage along the southeast coast of the island. Likewise, sites 62 to 70 on the east coast have limited stratigraphic spread. So as not to underestimate error, we bunched together sites that were stratigraphically close, arranging them into 11 groups spaced as evenly as possible through the sequence (Fig. 6). The mean of bedding normals then becomes $D,I=167.4°,71.8°$, $N=11$ (groups), $k=53$, $\alpha_{95}=6.3°$, which is our preferable estimate. Evidently, the section is a regularly inclined panel, with a mean dip of 18.2° (error 6.3°, $P=0.05$) towards 347.4°. Based on these field-observed bedding attitudes, the thickness of the sampled section is 1.2 km.

Directions of magnetization at each site are given in Table 1 and Fig. 7, and averages in Table 2. Site directions are well grouped and near vertical, relative to present horizontal. All except site 72 towards the top of the section are normally magnetized, so it appears that much of the sequence was deposited quite rapidly during a

Table 1
Data by sites

Site (polarity)	UTM(E,N)	M	n	s	$D_H{}^\circ, I_H{}^\circ$	k_s	$\alpha_{95}{}^\circ$
RI Ramsay Island lavas							
10(N)	339750,5824250*	1.32	6	13	002,88	195	3
11(N)	339775,5824250*	2.74	5	10	335,85	496	2
12(N)	339800,5824260*	1.23	6	11	308,86	180	3
13(N)	339850,5824300*	1.68	5	10	011,83	158	4
14(N)	339900,5824320*	1.21	5	9	289,88	215	4
15(N)	339950,5824330*	1.18	5	10	134,89	157	4
16(N)	339950,5824340*	1.71	5	9	233,86	120	5
17(N)	340070,5824500*	1.37	6	12	003,85	231	3
18(N)	340095,5824550*	2.59	5	9	110,82	377	3
37(N)	340100,5824600*	1.30	5	10	141,87	434	2
38(N)	340150,5824650*	1.71	5	10	337,87	448	2
41(N)	340200,5824660*	1.36	5	10	019,74	303	3
42(N)	340250,5824690*	2.14	5	10	359,85	158	4
43(N)	340300,5824750*	2.17	5	7	187,87	263	4
44(N)	340310,5824700*	2.55	5	7	317,87	230	4
45(N)	340370,5824760*	1.44	5	9	037,87	33	9
46(N)	340410,5824790*	1.96	5	8	327,84	51	8
47(N)	340495,5824800*	0.83	5	8	305,83	255	4
48(N)	340530,5824820*	2.21	5	8	343,81	194	5
49(N)	340565,5824840*	1.60	5	10	016,86	163	4
50(N)	340600,5824850*	1.37	5	10	324,87	297	3
60(N)	340700,5825550	1.24	5	10	358,82	415	2
61(N)	340950,5825850	1.04	5	10	357,75	173	4
62(N)	340850,5826460	1.21	6	12	003,77	152	4
63(N)	340870,5826330	2.29	5	9	032,70	288	3
64(N)	340930,5826370	1.49	5	10	062,66	188	4
65(N)	340900,5826580	0.65	5	10	219,78	282	3
66(N)	340820,5826520	2.72	6	11	341,77	486	2
67(N)	340900,5826580	3.11	5	10	293,86	540	2
68(N)	340710,5826760	4.94	5	10	323,69	400	2
69(N)	340780,5826730	5.28	5	10	332,72	229	3
70(N)	340820,5826630	2.86	7	14	320,72	471	2
71(N)	340850,5825170	0.75	9	18	004,80	378	2
72(R)	340360,5827430	0.17	7	14	063,−87	88	4
73(N)	340000,5827600	1.13	6	12	331,73	394	2
74(N)	339650,5828000	1.15	6	12	243,80	173	3
75(N)	339650,5828250	0.96	5	10	279,74	400	2
SL Sheila Lake Pluton							
01(N)	676300,5929500	0.82	6	12	067,78	199	3
02(N)	676400,5929400	0.72	6	12	023,74	236	3
03(N)	676700,5929000	0.35	5	10	031,80	146	4
04(N)	676400,5929500	0.14	6	12	010,78	29	8
05(N)	676300,5928800	1.10	5	10	002,74	152	4
06(N)	676400,5828900	0.69	5	10	024,81	107	5
07(N)	676200,5928700	0.08	5	10	068,81	405	2
08(N)	676400,5929300	0.34	5	10	014,81	164	4
09(N)	676200,5929700	0.43	6	12	015,72	116	4
10(N)	675900,5929600	0.54	6	11	067,70	157	4
KP Kano Pluton							
12(R)	666600,5906700	1.41	5	10	195,−42	292	3
13(?)	666500,5906900	1.50	6	9	168,28	92	5
14(R)	668200,5899600	0.29	5	9	202,−47	583	2
15(R)	667900,5899200	0.25	4	8	205,−52	91	6
16(R)	668500,5898900	0.89	5	10	204,−42	306	3
18(R)	668600,5898200	0.12	4	8	178,−54	154	5

Table 1 (*continued*).

Site (polarity)	UTM(E,N)	M	n	s	$D_H{}^\circ, I_H{}^\circ$	k_s	$\alpha_{95}{}^\circ$
LG Langara Island Pluton							
34(N)	630700,6013300	0.49	5	10	352,73	1559	1
35(N)	630400,6013600	0.53	5	10	005,72	730	2
36(N)	629900,6013700	0.42	5	10	359,69	844	2
37(N)	629200,6013600	0.50	5	10	001,68	1656	1
38(N)	628800,6013700	0.23	5	10	002,70	663	2
39(N)	627500,6013700	0.27	5	10	067,73	401	2
40(N)	626400,6013600	0.89	5	10	049,77	3495	1
41(N)	625500,6012200	0.37	5	10	056,73	74	6
42(N)	625400,6012500	0.22	5	10	053,71	1052	2
43(N)	626400,6013200	1.47	5	10	026,76	708	2
RS Rennell Sound dyke swarm							
21(N)	662700,5920800	2.96	5	10	031, 80	27	10
22(N)	660800,5921700	1.09	6	12	048, 82	20	10
23(N)	660100,5915300	0.91	6	11	328, 83	51	7
24(R)	654500,5928500	0.50	5	10	186, −81	400	2
25(R)	652800,5916300	0.41	5	10	188, −68	428	2
26(R)	655300,5915800	0.84	7	14	173, −80	159	3
28(N)	664500,5918700	1.03	6	12	022,77	21	10
29(N)	664300,5918800	0.76	6	11	143, 82	10	15
30(N)	665900,5916800	2.25	7	14	332,78	115	4
32(N)	664600,5915200	2.43	6	12	020,81	386	2
33(N)	664600,5920200	1.17	6	12	308,88	18	11
DB Dinan Bay lavas							
46(N)	653800,5949200	1.66	7	14	315,80	338	2
47(N)	654900,5947600	2.35	6	12	103,81	196	3
48(N)	654000,5946800	4.74	6	10	301,81	619	2
49(N)	654000,5947100	3.72	7	13	357,51	191	3
51(R)	653700,5946600	2.99	6	11	133, −76	101	5
52(R)	653500,5946600	2.13	7	14	133, −67	113	4
53(R)	653600,5947100	2.65	5	10	137, −71	242	3
54(R)	652200,5946300	1.29	6	12	100, −78	850	2
55(R)	652500,5946000	1.34	5	10	081, −86	237	3
56(R)	652800,5946100	1.77	6	9	156, −80	485	2
57(R)	653000,5949550	1.29	6	12	141, −78	1464	1
58(R)	653200,5945700	4.38	5	10	154, −57	100	5
59(R)	652900,5946200	4.62	3	6	157, −65	236	4
NW Cape Naden–Wiah Point sequence							
28(R)	5996300,659600	0.61	5	10	238, −66	1119	1
29(R)	5996000,658000	0.25	5	11	212, −54	281	3
30(R)	6002000,653200	1.68	4	8	252, −64	1290	2
31(R)	5976000,667750	1.20	4	6	234, −54	259	4
32(R)	5980000,677000*	1.44	5	8	236, −62	206	4
33(R)	5985000,676000*	1.79	4	8	232, −62	487	3
34(R)	5991000,675600*	3.04	4	8	242, −56	193	4
35(R)	5997500,675250	0.60	4	9	259, −63	850	2

UTM coordinates * interpolated; M intensity of magnetization (A m−1); n, s number of cores and specimens; $D_H{}^\circ, I_H{}^\circ$ declination and inclination of mean direction of magnetization of s specimens relative to present horizontal; k_s Fisher's precision; α_{95} radius of the circle of confidence ($P=0.05$). Rennell Sound dykes data obtained by line-fitting in the ranges 10–100mT and 400–600°C. Other data by step-cleaning, usually in the range 10–30mT or 500°C. The down dip direction and dip of bedding measured at Ramsay Island are site 10 353,11; 11 355,15; 12 015,14; 13 342,07; 14 005,08; 15 025,14; 16 016,21; 17* 016,21; 18 008,28; 37 357,32; 38 014,26; 41* 009,24; 42 004,22; 43 018,27; 44 024,25; 45 023,15; 46 020,23; 47 345,11; 48 351,20; 49 350,17; 50 025,16; 60 000,15; 61 014,14; 62 300,19; 63 276,34; 64 288,26; 65 331,22; 66 341,20; 67* 341,20; 68 315,20; 69* 315,20; 70 317,30; 71 318,18; 72 349,20; 73 012,27; 74 345,25; 75 345,34: * interpolated between sites above and below or taken directly from adjacent sites. Average down dip direction and dip of Dinan Bay lavas is 022°,10°, and of the Cape Naden–Wiah Point sequence is 325°, 10°.

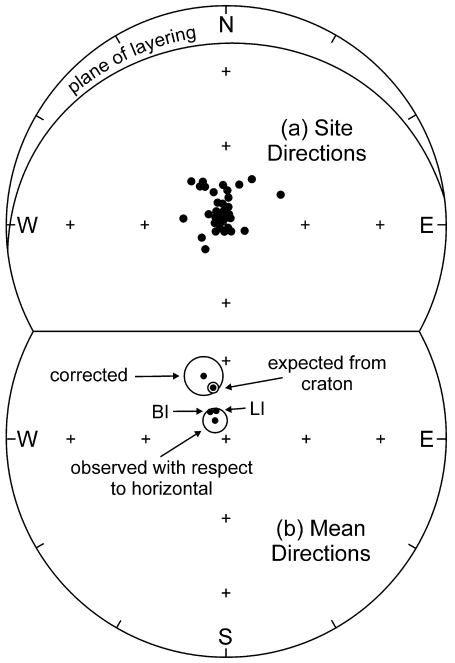


Fig. 7. Directions of magnetization from Ramsay Island. (a) Site mean directions with respect to present horizontal; site (72) with reversed polarity has been inverted. Mean plane of bedding is marked. (b) Mean of site directions with respect to horizontal and corrected for dip, compared with directions in two adjacent dyke swarms (BI, Burnaby Island swarm; LI, Lyell Island swarm (Irving et al., 1992)), and with the direction expected from observations from coeval rocks in continental North America (Table 3). The error in the mean is larger when corrected (7.9°) than when uncorrected (4.8°), because the former combines paleomagnetic and bedding (6.3°) errors.

single magnetochron, with site 72 representing a briefly reversed subchron.

We arranged the sites into 11 groups as before to ensure uniform coverage. Relative to present horizontal and irrespective of sign, the mean direction (334°,83°, $\alpha_{95}=5°$) is very close to, and not significantly different from, the mean directions of neighbouring dykes (Fig. 7), notably the Burnaby Island swarm (341°,80°, $\alpha_{95}=4°$) and the Lyell Island swarm (354°,81°, $\alpha_{95}=5°$). The magnetizations are not younger regional overprints because reversals are present.

The mean direction of the lavas, after correction for bedding, agrees with that expected from coeval flat-lying rocks of continental North America; the means differ by 5.1° which is not significant (Table 2, Fig. 7). Uncorrected, the direction disagrees with that expected. Hence, the magnetization of the lavas predates tilting.

The difference between observed and expected directions of magnetization gives an apparent tilt of the Ramsay Island sequence of $13.0°\pm5.0°$ towards 358.0° (Table 3). This is not significantly different from tilting indicated by bedding ($18.2°\pm6.3°$ towards 347.4°), and confirms the paleomagnetically determined tilt. We also conclude that oversteepening in the two adjacent dyke swarms which have statistically identical directions (Fig. 7) was caused by tilting.

In Table 2 (see footnote) the paleopole calculated from directions corrected for geologically determined tilts is given. This can serve as a Late Eocene to Early Oligocene reference paleopole for North America.

7. New results from Graham and Langara Islands

7.1. Sheila Lake Pluton

A U–Pb zircon age of 27 ± 0.3 Ma was determined by Anderson and Reichenbach (1991). Medium to fine grained diorite was sampled at 10 sites over 1 km north and east of Sheila Lake. Directions of magnetization are very well grouped within and between sites (Tables 1 and 2). They are directed steeply downward (Fig. 8), that is, they have normal polarity.

7.2. Kano Pluton

The U–Pb zircon age given by Anderson and Reichenbach (1991) is 32.2 ± 0.3 Ma. Medium to fine grained monzonite was sampled at eight sites, two 200 m apart on Van Inlet, and six on Dawson Inlet over a distance of 2 km. At two sites, magnetizations are entirely soft Brunhes overprints. Magnetizations at six sites are stable, showing excellent linear decay to the origin. Directions within sites are tightly grouped. Five sites have southwest directions with shallow negative inclinations: the polarity is reversed (Table 1, Fig. 8). One site has south-southeast directions and moderate positive inclination. Inclinations at all sites are

Table 2
Summary data[a]

	B	D_M°, I_M°	k_M	α_{95}°	λ°(N), ϕ°(E)
RI Ramsay Island lavas (52.5°N, 131.3° W)					
Direction hor. (N+R)	37(11)	333.7,83.1	92	4.8	
pole (horizontal)	11		27	9.0	64.0,215.2
Direction (stratigraphic)	37(11)	343.7,65.0	92	4.8	
Paleopole (stratigraphic)	11		41	7.0	78.1,119.9[b]
SL Sheila Lake Pluton (53.5°N, 132.3 W)					
Direction (N)	10	031.6,78.1	133	4.2	
pole	10		41	7.6	69.6,263.0
KP Kano Pluton (53.3°N, 132.5° W)					
direction (R)	5	017.3,47.8	80	8.6	
pole	5		57	10.2	62.7,13.7
LG Langara Island Pluton (54.3°N, 133.0° W)					
LGN direction (N)	5	000.0,70.6	979	2.4	
LGN pole	5		381	3.9	89.3,225.8
LGNE direction (N)	5	051.3,74.4	301	4.4	
LGNE pole	5		97	7.8	62.6,282.6
RS Rennell Sound dyke swarm (53.4°N, 131.5° W)					
direction (N+R)	11	010.4,81.9	103	4.5	
pole	11		30	8.5	68.4,236.0
DB Dinan Bay lavas (53.7°N, 132.7° W)					
direction hor. (N+R)	13	327.7,75.4	36	7.0	
pole (horizontal)	13		14	11.4	69.7,182.9
direction strat. (N+R)	13	348.8,68.1	36	7.0	
paleopole (stratigraphic)	13		19	9.8	83.3,137.8[b].
NW Cape Naden–Wiah Point sequence (54.1°N, 132.3° W)					
direction (horizontal, R)	8	237.2, −60.8	91	5.8	
pole (horizontal)	8		47	8.2	50.9,311.2
direction strat. (R)	8	219.7, −59.6	91	5.8	
paleopole (stratigraphic)	8		55	7.6	60.3,327.4[b]

[a] Directions and paleopoles relative to bedding are labelled 'stratigraphic'. Otherwise directions and poles are relative to present horizontal plane. Poles calculated as a mean of poles for each site (VGPs) quoted for the northern hemisphere poles. When both normal and reversed polarities are present, the directions are given irrespective of sign. B number of collecting sites. D_M°, I_M°, k_M, α_{95}, λ°(N), ϕ°(E) mean direction, precision, error ($P=0.05$), latitude and longitude of paleopole.

[b] Paleopoles calculated from directions corrected to geologically determined bedding: RI is a reliable reference paleopole for North America; DB, because of the less rigorous estimation of bedding, is an adequate but less reliable reference pole for North America; NW from the anomalous northwestern area of the archipelago (Fig. 11) has undergone clockwise rotation and cannot be used as a reference pole.

very different from the steep inclinations observed in other Tertiary rocks elsewhere in the Queen Charlotte Islands.

7.3. Langara Island Pluton

A U–Pb age of 26.8 ± 0.4 Ma was determined by Anderson and Reichenbach (1991). Polarities are all normal. Inclinations are intermediate to steep (Fig. 8). Five sites from east of Langara lighthouse have northerly directions, and five from around the lighthouse have northeast directions. The two groups are significantly different (Table 2).

7.4. Rennell Sound dyke swarm

Fifteen andesite and basalt dykes were sampled over the 12 km east–west extent of the swarm on Rennell Sound. The dykes, which are sub-vertical, strike approximately north-northeast to south-southwest. They range in width from 2 to 45 m,

Table 3
Tilt estimates

Rock unit	D_E°,I_E° (α_{95}°)	Tilt°azimuth° (error°)
SI, Selwyn Inlet dyke swarm (11)	353.6,69.1 (3.0)	15.3 at 323.2 (6.5)
TS, Tasu Sound dyke swarm (12)	349.5,68.0 (2.3)	10.9 at 000.2 (5.2)
LI, Lyell Island dyke swarm (12)	349.5,67.9 (2.3)	12.7 at 346.4 (5.5)
BI, Burnaby Island dyke swarm (13)	349.5,70.4 (2.0)	10.0 at 357.7 (7.7)
CB, Collison Bay dyke swarm (13)	349.5,70.4 (2.0)	07.4 at 354.9 (4.5)
RH, Rose Harbour dyke swarm (13)	349.6,70.3 (2.0)	12.0 at 010.0 (7.0)
MI, Masset Inlet volcanics (11)	353.5,69.6 (3.0)	16.0 at 329.1 (6.3)
LP, North Louise Island Pluton (13)	349.4,70.9 (2.0)	09.6 at 001.8 (4.4)
RI, Ramsay Island lavas (13)	349.5,70.5 (2.0)	13.0 at 358.0 (5.0)
SL, Sheila Lake Pluton (11)	353.5,69.5 (3.0)	13.5 at 319.4 (5.1)
RS, Rennell Sound dyke swarm (14)	351.5,70.3 (1.6)	12.3 at 338.9 (4.7)
DB, Dinan Bay lavas (11)	353.4,69.7 (2.9)	10.0 at 036.5 (7.6)

The observed mean directions (D_M,I_M) are given in Table 2 herein, and in table 4 of Irving et al. (1992). D_E,I_E, the expected mean directions and errors (α_{95}°), are calculated from mean craton paleopoles listed in table 5 of Irving et al. (1992) for the appropriate interval as numbered in the first column: (11) 10–30 Ma, (12) 20–40 Ma, (13) 30–50 Ma, 14 overall average 10–50 Ma. Errors ($P=0.05$) in the last column are for normals to bedding and equal to errors in the amount of tilt. Small corrections to table 5 of Irving et al. (1992) have been made.

generally between 4 and 7 m. There are no radiometric age determinations. Their age is considered to be in the range Late Eocene through Miocene (Souther and Jessop, 1991). Three dykes have only Bruhnes overprints, and one has lightning-type magnetization. As already noted, all directions have been calculated by line-fitting. Agreement within sites is variable and generally less good than for other units (Table 1). Three dykes are reversed, eight are normal, and have generally steep inclinations.

7.5. Dinan Bay lavas

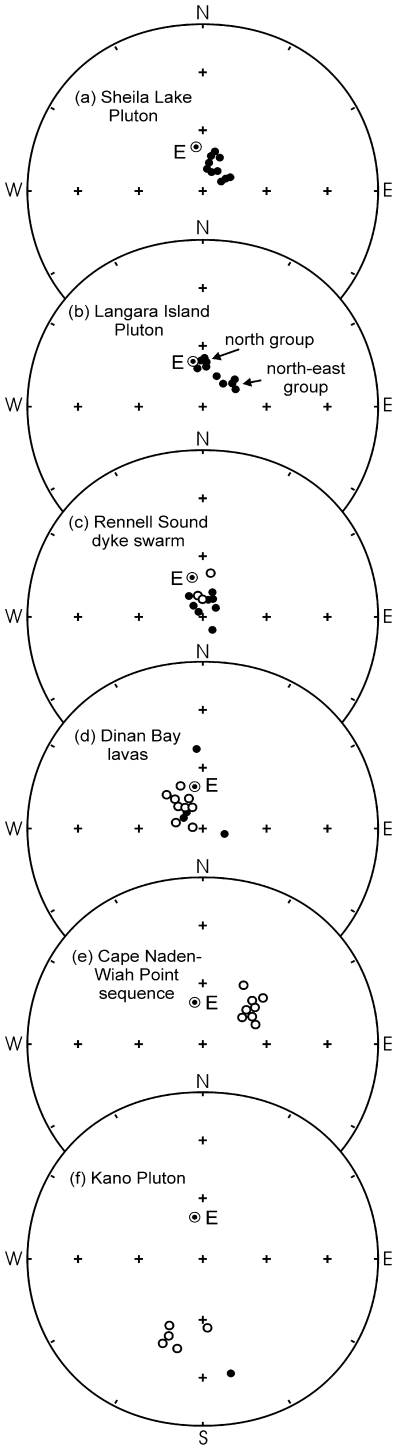
Massive, commonly amygdaloidal basalt was collected at 14 sites in separate flows, mainly to the south and southwest of the head of Dinan Bay. Three K–Ar ages from the area ranged from 24.4 to 18.9 Ma (Early Miocene), typical for lavas around Masset Inlet (Hickson, 1991; Hamilton and Dostal, 1993). The area is 15 km west-northwest of earlier sampling on Masset Inlet (Hicken and Irving, 1977). One site (50) has only a Brunhes overprint. Agreement among sites is good, and inclinations are generally steep. At outcrops, there are no accurate indicators of bedding, but from measurements of apparent dip of flows obtained by viewing hillsides with differing aspect, true dip was estimated to be down toward the north-northeast at 10°. In Table 2 (see footnote) the paleopole calculated from magnetization directions corrected for this geological estimate of tilt can serve as an Early Miocene reference paleopole for North America.

7.6. Cape Naden–Wiah Point sequence

Eight sites (five basalt flows, two dykes, one crystal tuff) were sampled in two sections on either side of the entrance to Naden Harbour. The section extends over 8 km, dips 10° down towards 325°, and has a thickness of 540 m. Basalt from the base of the sequence yielded a K–Ar age of 19.0 ± 0.7 Ma. Directions are west-southwest with moderate negative inclination and reversed polarity.

8. Discussion

The mean directions, relative to present horizontal, of the 15 rock units studied from MIC (four plutons, seven dyke swarms, and four lava sequences over a distance of 180 km) are plotted in Fig. 9. Ages range from Middle Eocene through Miocene. These results are based on data from



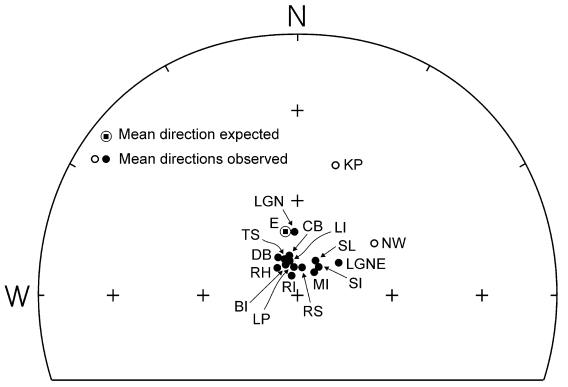


Fig. 9. Mean directions of magnetization of the Masset Igneous Complex, plotted on lower hemisphere with respect to present horizontal. Most rock units contain both normal and reversed polarities. All sites in KP and NW are reversed. Results are as follows: data of Hicken and Irving (1977), LP North-west Louise Island Pluton, and MI the first sampling of Masset Inlet lavas; data of Irving et al. (1992) BI Burnaby Island dyke swarm, CB Collinson Bay dyke swarm, LI Lyell Island dyke swarm, RH Rose Harbour dyke swarm, SI Selwyn Inlet dyke swarm, TS Tasu Sound dyke swarm; present study, DB Dinan Bay lavas, KP Kano Pluton (reversed direction inverted), Langara Island Pluton, LGN and LGNE, north and northeast directions respectively, NW Cape Naden–Wiah Point sequence, RI Ramsay Island lavas, RS Rennell Sound dyke swarm, SL Sheila Lake Pluton. E is the expected direction of the mean geomagnetic field with error ($P=0.05$) calculated from mean paleopole for North America for the interval 50 to 10 Ma (Fig. 10).

231 sampling sites, 1277 oriented cores, 2454 specimens. With two exceptions, rock units have anomalous magnetization directions that differ significantly from those expected from coeval rocks on continental North America. Twelve out of 15 form a cluster with mean inclinations at least 10° steeper than expected (TS, DB, CB, RH, BI, LI, LP, RI, RS, SL, SI, MI of Fig. 9) and corresponding poles that are strongly anomalous (Fig. 10). It is these that we consider first.

Fig. 8. Site mean directions of magnetization relative to present horizontal of each studied rock unit. In (a) to (e) solid dots and open symbols (reversed directions that have been inverted) are plotted on the lower hemisphere. In (f) directions with negative inclinations are plotted on the upper hemisphere as open symbols, and the single direction with positive inclination on the lower hemisphere as a solid symbol.

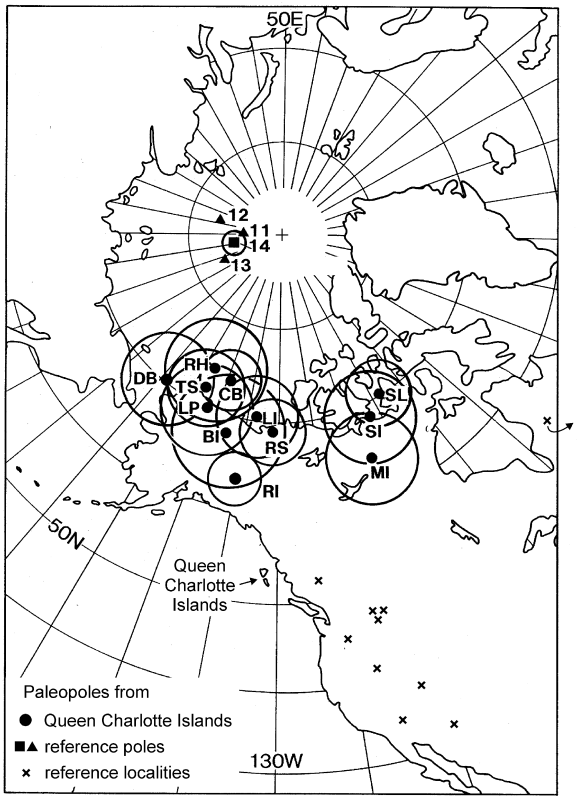


Fig. 10. Comparison of paleopoles from mid-Tertiary rocks of North America and from the oversteepened magnetizations of the Masset Igneous Complex. Paleopoles for North America are from Table 2 of Irving et al. (1992): 11, average for interval 10–30 Ma; 12, for 20–40 Ma, 13, 30–50 Ma, and 14 the overall average 10–50 Ma. Paleopoles from the Queen Charlotte Islands are labelled as in Fig. 9 and Table 3.

8.1. Oversteepened directions

Anomalous directions reflect one or more of the following causes: (1) tilting of the test locality, that is, rotation about a horizontal axis; (2) rotation of the test area about a vertical axis; (3) relative horizontal motion of the test and reference localities; (4) differing ages of test and reference locality rocks; (5) inaccurate recording of the geomagnetic field by test locality rocks; (6) the time averaged geomagnetic field differs markedly from that of a geocentric dipole. First we consider possible cause (6).

For the interval 50–10 Ma, paleopoles from the North American craton are in excellent agreement – averages for time windows within this range do not differ significantly (Fig. 10), and their precisions are uniformly high ($K=271$–676, documented in Irving et al., 1992). Besse and Courtillot (1991) have shown that for the same interval there is also excellent agreement among North American, Eurasian (impact breccia from the Ries meteorite crater, volcanics from the Vogelsberg in Germany, and Lower Silesian in Poland), and African (volcanics of Libya and the East African Rift in Kenya, Tanzania, and Ethiopia) cratonic paleopoles when corrected for plate motions; averages for time windows within this range do not differ significantly from one another, and precisions are again very high (K ranges from 130 to 244, Besse and Courtillot (1991), Table 6). The localities from which these data were obtained are globally widespread. Hence the time averaged geomagnetic field during the deposition of the MIC approximates very closely that of a geocentric dipole; cause (6) is not responsible for the anomalous results from the Queen Charlotte Islands.

Because inclinations are steep, rotations (cause (2)) cannot be responsible. Eocene rocks from the western element of the Cordillera, situated to the south of the Queen Charlotte Islands on Vancouver Island (Irving and Brandon, 1990) and to the north in Alaska (Hillhouse and Gromme, 1982; Hillhouse et al., 1985) have magnetic inclinations, and therefore paleolatitudes, concordant with those of mainland North America, so the islands were essentially in place latitudinally relative to North America by the mid-Tertiary. Any longitudinal shift that they might have undergone would not have affected directions of magnetization. Hence, cause (3) is unlikely. Cause (4) also is unlikely because the North American reference field did not change significantly from Late Eocene through Miocene, the interval in question (Diehl et al., 1983, analysis in Irving et al., 1992). Finally, cause (5) is improbable, because laboratory tests on a variety of different lithologies of the MIC have shown that remanent magnetization is acquired parallel to the ambient field (Irving et al., 1992). Therefore tilting is the most likely cause of oversteepening.

Our Ramsay Island test shows that over-steep-

ening, which is the same as that in adjacent dykes (Fig. 7), was caused by tilting to the north. The same is true also, with less certainty, of the Dinan Bay lavas, where correction for the inferred dip of 10° to the north-northeast brings the mean direction of magnetization relative to present horizontal (327.7°,75.4°, $\alpha_{95}=7.0°$), into better agreement (348.8°,68.0°) with that expected 353.4°,69.7°, $\alpha_{95}=2.9°$ (Table 2).

All evidence indicates that the general oversteepening of magnetizations (Fig. 9) reflects northerly tilting over much of the archipelago. The tilts required to bring magnetizations into agreement with those from untilted rocks of continental North America are given in Table 3 and plotted in Fig. 11. The average of bedding poles is 168.5°,78.9°, $k=261$, $\alpha_{95}=2.7°$, corresponding to an average tilt of 11.1° down towards 348.5°. Tilting cannot have occurred as a single block, otherwise upper mantle rocks corresponding to depths of 34 km would be exposed in the south, and they are not. It seems that the archipelago has been segmented into many panels. Assuming planar normal faults dipping at 60°, the total extension would have been about 10%. We regard this as a minimum estimate.

As a final check we have repeated the tilt calculations using the globally derived reference paleopole calculated by Besse and Courtillot (1991) to obtain the reference fields. Individual tilts, expressed as bedding poles, yield a mean of $D,I=155.8°,78.9°$, $k=272$, $\alpha_{95}=2.5°$, which corresponds to a mean tilt for the Queen Charlotte Islands of 11.1° towards 335.8°. The amount of tilt is the same as that determined from North American data alone. The difference of 13° in the direction of tilt may seem to be large, but it is not significant, only reflecting the difficulty in accurately determining azimuth when tilt is low. The incorporation into the reference field of data from coeval rocks from other continents corrected for plate motions confirms that the oversteepening of Queen Charlotte Island Tertiary paleomagnetic data is real, and that tilting is the only plausible cause.

8.2. Age of tilting

At tilted localities tilting must have post-dated emplacement of the MIC. The youngest rocks affected are the Middle Miocene Selwyn Inlet dyke swarm (13 ± 6 Ma), and their tilt is the same as that in older dykes. This broad similarity in the direction and magnitude of tilts (Fig. 11) is best explained by assuming that there was a single north–south extension event. Extension (and tilting) was likely to have ceased with the initiation of oblique convergence in the later Miocene or earliest Pliocene (Fig. 4). Hence tilting was most probably Middle and/or Late Miocene. In sharp contrast to older north–south trending dykes, the Selwyn dykes trend east–west (Fig. 2) implying north–south extension, which is consistent with their northerly tilt. Hence, north–south extension and concurrent northerly tilting probably occurred soon after the intrusion of the Selwyn dykes and long after formation of most of the MIC; tilting was not associated with the main igneous activity.

This argument is based partly on dating of the Selwyn Inlet swarm, and could be criticized, because there is only a single age determination with large error. One could argue that tilting occurred as a result of oblique convergence during the Pliocene by the mechanism proposed by McCaffrey (1996). However, tilting is much more likely to have occurred during the Middle through Late Miocene, because there is no evidence of extension during the Pliocene, volcanism and deposition of rift-related sediments having long ceased.

8.3. Directions that are not oversteepened

Results from the Naden Bow Valley bore-core from northwestern Graham Island that are not oversteepened (Wynne and Hamilton (1989); NH in Figs. 3 and 11). This vertical, 1148 m, core is in Lower Miocene lavas (K–Ar age 23 Ma), and detailed sampling revealed two reversals. The mean inclination ($69°\pm3°$) agrees closely with that expected (70°). Tilting in the immediate neighbourhood therefore, is unlikely.

Other magnetizations described herein also are not oversteepened. There are two significantly different groups of directions in the Langara pluton. One, LGN, has a mean direction in agreement with that expected, the other, LGNE, has an inclination close to expected, but a declina-

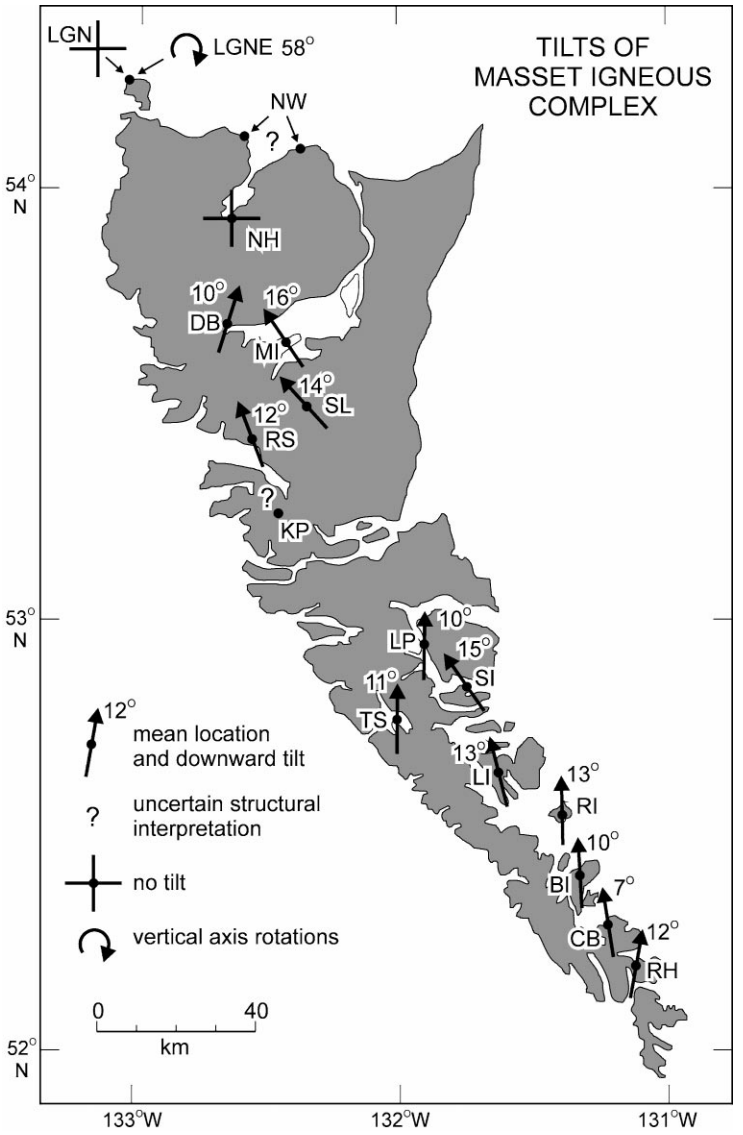


Fig. 11. Structural interpretation of paleomagnetic results from the Masset Igneous Complex. Rock units labelled as in Fig. 9. Inferred tilts are listed in Table 3. The average tilt is 11.1° down towards 348.5°.

tion rotated 58° clockwise. The presence of these two groups presumably reflects local faulting within the pluton. Anomalous directions from the Cape Naden–Wiah Point succession have a mean inclination that is marginally shallower ($61\pm6^\circ$) than expected ($70\pm2^\circ$), and declinations that marked clockwise deviation ($47\pm17^\circ$). These are so whether or not corrections for tilt are applied. Tilting, with or without rotation, could have been responsible, and the question mark at NW in Fig. 11 expresses that ambiguity. However, the barely significant shallowing of inclination contrast with the strongly anomalous declination, and indicates that there has been substantial clockwise rotation of the Cape Naden–Point Wiah succession (Figs. 8 and 9).

These three results (NW, LG, NH), which do not conform to the general pattern of oversteepening, may delineate a structural domain, adjacent to Dixon Entrance, distinct from the rest of the Queen Charlotte Islands.

The Kano pluton in central Graham Island yields inclinations that are shallower than expected, and very different from those elsewhere in the Queen Charlotte Island (Fig. 9), but data are few, areal coverage limited, and we make no structural interpretation.

8.4. Comparisons with the geomagnetic polarity time scale of Cande and Kent (1992)

These studies of the MIC provide five dated polarities, all of which are consistent with the polarity time scale of Cande and Kent (1992). The Northwest Louise Island pluton (U/Pb zircon age 34.3 ± 0.6 Ma, Anderson and Riechenbach, 1991) yielded normal polarity (Irving et al., 1992), consistent, within error, with Early Oligocene chron 15n. The reversely magnetized Kano pluton (U/Pb zircon age 32.2 ± 0.3 Ma, Anderson and Riechenbach, 1991) falls in mid-Oligocene chron 12r. The normally magnetized Sheila Lake and Langara plutons (U/Pb ages 27.0 ± 0.3 Ma and 26.8 ± 0.4 Ma respectively, Anderson and Riechenbach, 1991) fall, within error, in Late Oligocene chron 9n. The Cape Naden–Wiah point volcanic sequence (19.0 ± 0.7) falls, within error, in Early Miocene chron 5Er. This consistency testifies to the reliability of the age and magnetism results from the MIC.

9. Evolution of Queen Charlotte Basin

Prior to the Middle Eocene, the tectonic regime was one of convergence of an oceanic plate subducting beneath North America. Since then the Queen Charlotte Basin has had, at its western margin, a predominantly transform fault, the Queen Charlotte Fault.

Except for the past few million years, for which a tranpressional regime is well documented, plate reconstructions are not precise enough to provide a detailed timetable of the variations in the departures from purely transform motion. Notwithstanding, the following model can be formulated, taking into account the structure of offshore basins, and the structure, petrology, geochemistry and paleomagnetism of the MIC. The model is derived from the ideas of many workers, and is based on the notion that plate motions evolved from highly oblique divergence during the Middle Eocene through Early Miocene (transtensional regime), to purely transform sometime during the Middle through Late Miocene, to highly oblique convergence (transpressional regime) since the latest Miocene (Figs. 2 and 12).

9.1. Phase 1, transtension

Hyndman and Hamilton (1993) proposed that it was the large change in the direction of plate motion (Fig. 4) from strong compression to transtension during the Eocene (Engebretson et al., 1985; Stock and Molnar, 1988) that initiated the formation of the Queen Charlotte Basin. The accompanying large decrease in relative plate velocities also may have assisted the establishment of this transtensional regime. Transtension continued for about 25 Ma, during which the main part of the MIC and associated sediments (Unit 1 of Higgs, 1991) were deposited in diachronous rift successions. The sense of extension is essentially east–west, as evidenced by the general north–south trend of dykes (Souther and Jessop, 1991, Fig. 2), and by the predominantly north–south orientation of the extensional faults in Hecate Strait and Queen Charlotte Sound (Rohr and Dietrich, 1992; Rohr and Currie, 1997). The initial width of the basin may have been about two thirds of present (Hyndman and Hamilton, 1993).

9.2. Phase 2, transform margin

In the Middle and/or Late Miocene, plate motions became more purely transform (Fig. 12). East–west extension ceased. The basin began to subside, and sediments of Higgs's Unit 2 accumulated. Igneous activity decreased and finally ceased in the Late Miocene. Incipient strike-slip faulting

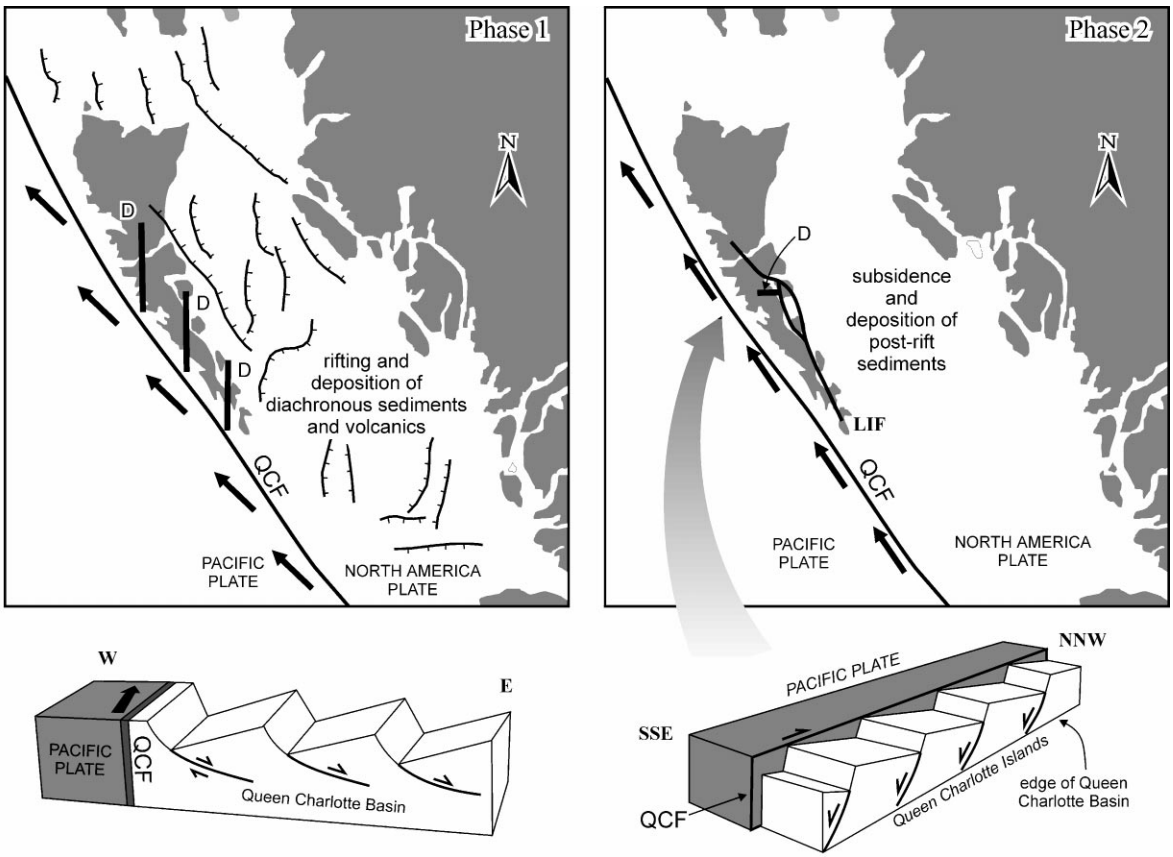


Fig. 12. Model of three phase evolution of Queen Charlotte Basin. Phase 1: transtension, Middle Eocene through Early Miocene. Normal faults shown schematically from Rohr and Dietrich (1992). D, predominant orientations of dyke swarms (Souther and Jessop, 1991). Initial width of the basin may have been about two-third of the present (Hyndman and Hamilton, 1993). Phase 2: transform motion, Middle through Late Miocene. Marginal slivers extended by minimum of 10% and tilted, intrusion of east–west Selwyn Inlet dyke swarm, D. LIF, the Louscoone Inlet transcurrent fault system. For Phase 3: transpression, Pliocene to Present, see Figs. 1 and 2.

(the Louscoone fault system of Lewis and Ross (1991) could have been such a fault) activated by the transform fault, affected the western edge of the basin, and the resulting marginal slivers were extended by about 10% or more, fragmented and tilted (Irving et al., 1992). The youngest dyke swarm, the east–west, Middle Miocene, Selwyn Inlet swarm, was intruded during early phase 2.

9.3. Phase 3, transpression

In the latest Miocene or earliest Pliocene (Figs. 1 and 2), plate motion became highly oblique convergent (Engebretson et al., 1985; Stock and Molnar, 1988), resulting in the uplift and erosion of the Queen Charlotte Islands (Yorath and Hyndman, 1983). Elsewhere, subsidence of the basin and deposition of sediments of Unit 2 continued.

In the spirit of free speculation of which Myrl Beck is such a notable exemplar, let us look say 40 Ma ahead to when the Queen Charlotte Basin has become intimately incorporated into the Western Cordillera as just another tectonostratigraphic terrane, the Masset Terrane. The internal basinal structure of the basin has been preserved,

but its present boundaries, in particular the Queen Charlotte Fault, have been replaced by much younger faults of different character. We can ask, 'what tectonic regime would future geologists infer for this Masset Terrane?' With only the internal structure retained, their confident answer is likely be, 'an extensional basin with an early phase approximately at right angle to the second', and this may then be carried forward as a major building block in regional tectonic analysis. But this conclusion would be correct only in a very local sense. Regionally, the Masset Terrane was created by transform motion on a major plate boundary; the overall tectonic regime is transform not extensional. It would be a case of not seeing the wood for the trees.

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